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MASTER THESIS



论文题目 5G 移动通信基站天线关键技术及其
特征模分析方法研究

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摘 要

本文档使用 UESTC Thesis 模板编译生成，用于展示电子科技大学学位论文的基本结构与排版效果。

本文档涵盖中英文摘要、章节层次、数学公式、图表、文献引用、主要符号表、缩略词表、附录及攻读学位期间取得的成果等常用内容。文中的文字、数据和图表仅供排版演示，使用者可根据实际论文内容进行修改或替换。

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关键词：电子科技大学；学位论文；LaTeX 模板；论文排版

ABSTRACT

This document is compiled using the UESTC Thesis template to demonstrate the basic structure and typesetting of dissertations and theses at the University of Electronic Science and Technology of China.

It covers Chinese and English abstracts, heading levels, mathematical equations, figures, tables, citations, lists of symbols and abbreviations, appendices, and academic achievements during the degree program. The text, data, and figures are provided solely for typesetting demonstration and may be modified or replaced according to the user's thesis.

The organization and writing conventions of dissertations and theses vary across disciplines. Users may consult outstanding dissertations and theses in their fields through academic databases such as CNKI and Wanfang Data to learn common approaches to organization and scholarly expression. Thesis formatting should follow the latest writing standards and related templates published by the UESTC Graduate School, available at <https://gr.uestc.edu.cn/xiazai/114/3917>. Based on these official documents and Word templates, the UESTC Thesis template reproduces their principal structure and layout with high fidelity and can be used confidently for thesis writing and typesetting. Before final submission, users should nevertheless confirm any additional requirements issued by their school or department.

Keywords: UESTC; Dissertations and Theses; LaTeX Template; Thesis Typesetting

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Symbols

Symbol	Description	Page
π	The circle constant	2
θ	A scalar parameter	2
$\mathbf{x}$	An input vector	2
$\mathbf{y}$	An output vector	2
$\mathbf{A}$	A transformation matrix	2

Acronyms

Abbreviation	Full Name
AI	Artificial Intelligence
UESTC	University of Electronic Science and Technology of China

Chapter 1 Template Usage Example

This document briefly introduces the main features of the template. For more detailed instructions, see the Markdown user guide, `GUIDE.md`, in the project directory.

1.1 Document Configuration

Edit `main.tex` to set the degree type, title, author information, and other options:

- Degree type: use `master` for a master's thesis and `doctor` for a doctoral dissertation;
- Thesis language: download the Chinese or English package; each package uses `main.tex`.

Import the cover, Chinese and English title pages, and declaration page as follows:

1. Download the appropriate Word template: <https://gr.uestc.edu.cn/xiazai/114/3917>
2. Enter the thesis information;
3. Export the cover and Chinese and English title pages as `cover.pdf`, then replace the file of the same name in the project. To replace the declaration of originality and authorization page, scan the signed page as `declaration.pdf` and replace the corresponding project file.

1.2 Thesis Structure

A dissertation/thesis consists of preliminary pages, the main text, and final pages. Their components and order are shown in Table 1-1.

1.3 Second-Level Heading

1.3.1 Third-Level Heading

1.3.1.1 Fourth-Level Heading

Regular text, **bold text**, *italic text*, and underlined text.

Table 1-1 Basic Structure of a Degree Dissertation/Thesis

Component	Contents	Component	Contents
Preliminary Pages	1. Cover Page	Main Text	1. Introduction (Chapter 1)
	2. Chinese and English Title Page		2. Body of Dissertation/Thesis
	3. Statement and Authorization		3. Conclusion (last Chapter)
	4. Abstract in Chinese and English	Final Pages	1. Acknowledgements
	5. Contents		2. References
	6. Figures and tables (if necessary)		3. Appendix (if necessary)
	7. Symbols, Acronyms, etc. (if necessary)		4. Research Results

Note: Components marked “if necessary” may be included according to the thesis and disciplinary conventions. To omit a component, comment out its input command in `main.tex`. For example, omit the list of symbols with `%\input{chapters/symbols}`.

1.4 Mathematical Expressions

Let the constant π and scalar θ satisfy $\theta = \pi/2$. A vector $\mathbf{x}$ is transformed by a matrix $\mathbf{A}$ into a vector $\mathbf{y}$, as shown in (1-1).

$$\mathbf{y} = \mathbf{A}\mathbf{x}. \quad (1-1)$$

Here, $\mathbf{y}$ is the output vector, $\mathbf{A}$ is the transformation matrix, and $\mathbf{x}$ is the input vector.

1.5 Theorems and Proofs

Definition 1.1 A matrix $\mathbf{A}$ is orthogonal if $\mathbf{A}^\top \mathbf{A} = \mathbf{I}$.

Theorem 1.2 An orthogonal matrix preserves the Euclidean norm of a vector.

Proof Since $\mathbf{A}^\top \mathbf{A} = \mathbf{I}$, $\|\mathbf{A}\mathbf{x}\|_2^2 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|_2^2$. ■

1.6 Figures

Figure 1-1 shows the UESTC visual identity.^①

① When a cross-reference to a figure, table, or theorem is followed by text, retain a normal space after the command, for example between `\autoref{fig:uestc-logo}` and “shows.” Omit the space before punctuation.

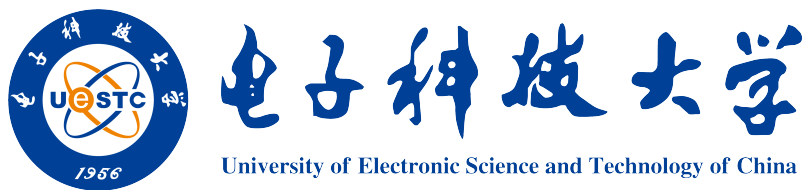


Figure 1-1 UESTC Visual Identity

Note: This sentence demonstrates the figure-note style.

1.7 Citations and Cross-References

Table 1-1 demonstrates the three-line table and table-note styles. Common source types include journal articles^[1], international conference papers^[2], books^[3], and arXiv preprints^[4].

Acknowledgements

This template was created by Xovee Xu.

GitHub repository: <https://github.com/Xovee/uestc-thesis>

For questions or suggestions, please email: xovee.xu@gmail.com

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Appendix A Supplementary Material

Appendix B Experimental Results

Research Results Obtained During the Study for Master's Degree

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